Notes: Caballero and Krishnamurthy (JF, 2008)
Collective Risk Management in a Flight to Quality Episode

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1 Big picture

- **Question.** Why do severe flight-to-quality episodes create economy-wide disengagement from risky activities, liquidity hoarding, and immobility of capital across markets? What can a lender of last resort do in such episodes?
- **Core answer.** The paper argues that some crises are not just high-risk states. They are *uncertainty* states in which agents do not trust their probability models. Agents use worst-case reasoning, overprotect themselves individually, and collectively waste scarce liquidity.
- **Key ingredients.** The model combines two frictions: limited aggregate liquidity and Knightian uncertainty about how aggregate shocks map into each agent's own liquidity needs.
- **Central mechanism.** When liquidity is scarce, cross-insurance is valuable: agents hit early by shocks should receive more resources because first shocks are more likely than second shocks. But under Knightian uncertainty, each agent worries that he will be hit late, after others have already used the liquidity pool. This shifts the allocation toward self-insurance and reduces efficient cross-insurance.
- **Flight-to-quality interpretation.** Agents respond to uncertainty by demanding safe, noncontingent, liquid claims. Intermediaries and trading desks hoard capital. Risk capital becomes locked into local markets rather than moving to where it is socially most valuable.
- **Collective bias.** Each individual agent protects against a private worst-case scenario. But the collection of all private worst-case scenarios is internally inconsistent: everyone cannot be late relative to everyone else. This creates a macroeconomic distortion that no individual internalizes.
- **Policy result.** A central bank or lender of last resort can improve outcomes by committing resources only in extreme low-probability states. This credible promise changes private ex ante portfolios and frees private liquidity for more likely states.
- **Public-private complementarity.** In standard moral-hazard stories, public insurance crowds out private insurance. In this paper, properly designed last-resort insurance complements private insurance because it reduces worst-case anxiety and encourages agents to reallocate private liquidity toward more likely shocks.
- **Timing matters.** Early, frequent, or non-last-resort interventions can worsen incentives. The policy has value because it is reserved for extreme tail states in which private liquidity is exhausted.
- **Economic takeaway.** The paper provides a theory of central-bank intervention when crises are driven by Knightian uncertainty rather than only by liquidation externalities or multiple equilibria.

2 Incremental contribution

- **Knightian uncertainty as a crisis primitive.** The paper adds model uncertainty to the financial-crisis toolkit. Earlier crisis models emphasized liquidation externalities, fire sales, or coordination failure. Caballero and Krishnamurthy instead formalize the idea that new, untested shocks cause agents to distrust their models and act on worst-case beliefs.
- **A collective-risk-management externality.** The paper identifies a distinctive externality: agents' individual robust decisions are not collectively coherent. Each agent behaves as if he might be last in line for liquidity, but the economy cannot have every agent be last. The social planner averages across agents and sees that aggregate liquidity is wasted.
- **A new role for the lender of last resort.** The lender of last resort is not valuable only because it prevents runs or provides missing liquidity. It is valuable because a credible promise to supply liquidity in rare worst-case states changes private behavior in common states.
- **Public and private insurance as complements.** A major contribution is the reversal of the standard moral-hazard intuition. When private agents are over-insuring tail states, public insurance of those tail states can induce private agents to insure more likely shocks, increasing the efficiency of private insurance.

- **A mechanism for market-wide capital immobility.** The paper explains why capital may fail to move across markets even when some markets are unrelated to the initial shock. Uncertainty about future liquidity needs causes capital to be locked up as precautionary liquidity rather than deployed where returns are high.
- **A bridge from decision theory to crisis policy.** The model imports max-min expected utility and robustness concerns into a macro-financial crisis setting and connects them to practical central-bank concerns about liquidity injection, payments-gridlock risk, and systemic risk management.
- **Policy guidance for financial innovation.** The paper highlights that crises are likely around new instruments or new market structures because agents lack tested models of how shocks propagate through them. The policy message is not simply "intervene when prices fall," but "intervene when new uncertainty causes inefficient hoarding."

3 Model objects and measurement

3.1 Agents, liquidity shocks, and dates

- The economy is studied conditional on being in a turmoil period.
- There is a continuum of competitive agents indexed by $\omega \in [0, 1]$.
- Each agent has an initial liquidity endowment Z.
- Agents may receive liquidity shocks requiring immediate consumption or capital.
- The economy can experience zero, one, or two waves of shocks.
- In a wave, one-half of agents are affected.
- The probability of at least one wave is $\phi(1)$; the probability of two waves is $\phi(2)$.
- Utility from shock-time consumption is concave, $u(c)$; utility from terminal consumption is linear with discount factor or marginal utility β.

Interpretation: The agents can be read literally as financial intermediaries or as trading desks, hedge funds, market makers, corporations with credit lines, or banks in a payment system. The model abstracts from detailed institutional structure to isolate a liquidity allocation problem.

3.2 Consumption plan and event states

Let s denote the state for an individual agent: the number of aggregate shock waves and whether the agent is hit first, hit second, or not hit. A consumption plan is

$$C = \{C^s\}_s.$$

For example, $s = (2, 1)$ means two waves occur and agent ω is hit in the first wave; $s = (2, 2)$ means two waves occur and the agent is hit in the second wave.

Interpretation: The key distinction is whether a given agent is early or late in the sequence of liquidity shocks. When aggregate liquidity is scarce, the efficient allocation favors earlier and more likely liquidity needs.

3.3 Aggregate versus individual probabilities

Agents know aggregate probabilities $\phi(1)$ and $\phi(2)$, but are unsure about their own relative likelihood of being hit first or second. The paper writes the perceived probability distortion as

$$\theta_\omega(2) - \frac{\phi(2)}{2},$$

with support roughly bounded by a parameter K.

- $K = 0$ is the benchmark risk case: agents share the same precise probabilities.
- $K > 0$ is the robustness or Knightian-uncertainty case: each agent considers a set of possible probability models.
- Larger K means more uncertainty about whether the agent is early or late in the shock sequence.

Interpretation: Agents are not uncertain about the aggregate probability of a crisis. They are uncertain about their own place in the queue for liquidity. This is what creates the conflict between individual worst-case thinking and collective feasibility.

3.4 Knightian uncertainty and max-min utility

Agents choose portfolios or liquidity allocations using a max-min rule:

$$\max_{C} \min_{\theta_\omega \in \Theta} \sum_{s} p^{s,\omega}(\theta_\omega) U(C^s).$$

Interpretation: The max-min rule captures robust risk management. The agent chooses the allocation that performs best under the worst probability model in the set of plausible models.

3.5 Sufficient versus insufficient aggregate liquidity

Define

$$c^* = u'^{-1}(\beta).$$

If $Z > c^*$, liquidity is sufficient; if $Z < c^*$, liquidity is insufficient.

- With sufficient liquidity, first- and second-wave agents can both receive c^*.
- With insufficient liquidity, not every shock can be fully insured.
- Flight-to-quality behavior and lender-of-last-resort value arise only in the insufficient-liquidity case.

Interpretation: Scarcity is essential. Knightian uncertainty matters most when agents cannot all be fully insured against every possible liquidity shock.

4 Models and theoretical specifications

4.1 Benchmark economy with $K = 0$

When agents have precise probabilities, the planner maximizes expected utility subject to resource constraints. In the insufficient-liquidity case, the reduced problem is

$$\max_{c_1} \ \phi(1)u(c_1) + \phi(2)u(2Z - c_1) + [\phi(1) - \phi(2)]\beta(2Z - c_1).$$

The first-order condition implies

$$\beta < u'(c_1) < u'(2Z - c_1),$$

so

$$c^* > c_1 > Z > c_2.$$

Interpretation: With scarce liquidity, first-wave agents get more insurance than second-wave agents. This is efficient because first-wave shocks are more likely: a one-wave event can stop after the first wave, but a second wave happens only conditional on the upper branch.

4.2 Proposition 1: benchmark allocation

Result block: In the $K = 0$ benchmark:

- If $Z < c^*$, the economy has insufficient aggregate liquidity. Agents are partially insured, and first-wave shocks receive more liquidity than second-wave shocks: $c^* > c_1 > Z > c_2$.
- If $Z > c^*$, the economy has sufficient aggregate liquidity. Agents are fully insured: $c_1 = c_2 = c^*$.

Economic message: The benchmark clarifies what efficient cross-insurance looks like. Scarce liquidity should be allocated more to earlier and more likely liquidity needs.

4.3 Implementation: intermediation or contingent claims

The benchmark allocation can be implemented in two ways:

- **Financial intermediation.** Agents deposit Z into an intermediary and receive state-contingent withdrawal rights.
- **Contingent claims.** Agents trade claims that pay when the agent receives a shock.

Interpretation: This dual implementation is important. It means the paper is not only about banks. It applies to trading desks, hedge funds, commercial paper markets, credit lines, and payments systems.

4.4 Robust economy with $K > 0$

When agents are Knightian, the planner's objective becomes

$$\max_{C} \min_{\theta_\omega \in \Theta} \sum_{s} p^{s,\omega}(\theta_\omega) U(C^s).$$

The key worst-case distortion is that an agent increases the perceived likelihood of being hit second.

For insufficient liquidity, the first-order condition can be written conceptually as

$$\text{worst-case weight on first shock} \times u'(c_1) = \text{worst-case weight on second shock} \times u'(c_2) + \text{terminal-value term}.$$

Interpretation: Knightian uncertainty pushes agents toward making c_2 closer to c_1. They insure late shocks more than is socially efficient because each agent fears being late.

4.5 Proposition 2: partial and full robustness

Result block: Let $\bar{K}$ be the threshold uncertainty level. In the insufficient-liquidity case:

- For $0 < K < \bar{K}$, agents choose a partially robust allocation with $c_2 < Z < c_1 < c^*$; as K rises, c_1 falls and c_2 rises.
- For $K > \bar{K}$, agents choose a fully robust allocation: $c_1 = Z = c_2 < c^*$.
- If $Z > c^*$, uncertainty has no effect on the full-insurance allocation $c_1 = c_2 = c^* < Z$.

Economic message: As model uncertainty rises, agents stop cross-insuring efficiently. In the fully robust case, each agent effectively self-insures with his own endowment Z.

4.6 Flight to quality as a comparative static in K

A flight-to-quality episode is a move from $K = 0$ to $K > 0$ after an unusual event. Examples include Russia/LTCM in 1998, credit-line drawdowns, or interbank payments-gridlock concerns.

- Agents rewrite contracts to protect themselves against the possibility of being late.
- Trading desks hoard committed capital rather than allowing a central risk manager to allocate risk capital flexibly.
- Firms draw credit lines preemptively.
- Banks hoard liquid securities rather than providing interbank liquidity.

Interpretation: Flight to quality is not simply a price movement. It is an allocation shift away from contingent risk sharing and toward safe, liquid, individually controlled resources.

4.7 Central bank objective and aggregation

The central bank's objective is ex post aggregate utility:

$$V^{CB} = \int_{\omega} \sum_{s} p^{s,\omega} U(C^s) d\omega = \sum_{s} p^s U(C^s).$$

Interpretation: The individual Knightian distortions disappear when averaged across agents. The central bank does not need more information than private agents; it needs to evaluate the collective allocation rather than each agent's private worst case.

4.8 Proposition 3: collective bias and value of reallocation

Starting from the robust equilibrium, consider increasing c_1 and reducing c_2 and terminal consumption. The paper shows that the central bank gain is positive when $K > 0$ and $Z < c^*$.

Result block: For any $K > 0$, if liquidity is insufficient, agents choose too much insurance against being hit second relative to being hit first. A central bank maximizing aggregate ex post utility can improve outcomes by reallocating insurance toward the first shock.

Economic message: The central bank corrects a collective overconservatism. The inefficiency is not due to private information or coordination failure; it is due to inconsistent individual robust beliefs.

4.9 Lender of last resort policy

The lender of last resort promises resources in a tail event. Let dZ be public insurance supplied in the second-shock event. Private agents then reoptimize. The total value of the intervention has two components:

$$V^{CB,total} = V^{CB,direct} + \text{portfolio reoptimization gain.}$$

Interpretation: The direct benefit is the resources supplied in the tail state. The indirect benefit is that private agents reduce inefficient tail-state self-insurance and increase insurance against the more likely first shock.

4.10 Proposition 4: public-private complementarity

Result block: For $K > 0$ and $Z < c^*$, the total value of a lender-of-last-resort policy exceeds its direct value:

$$V_{Z_G}^{CB,total} > V_{Z_G}^{CB,direct}.$$

Economic message: Public insurance is unusually powerful because it unlocks private liquidity. The central bank may rarely disburse resources, but its credible commitment changes private allocations before the tail event occurs.

4.11 Why the intervention must be last resort

If the central bank intervenes too early, private agents may shift resources toward later shocks in a way that worsens the original overconservatism. The value of the policy is highest when it is directed at extreme and unlikely states.

Interpretation: This gives the paper a nuanced policy recommendation: do not insure common shocks; credibly insure the extreme states that generate worst-case anxiety.

5 Theoretical design and identification logic

5.1 What is identified by the model

The paper is theoretical, so it does not estimate parameters from data. Its identification logic is conceptual: it separates an *uncertainty shock* from a standard liquidation or coordination shock.

- A liquidation shock reduces resources or collateral.
- An uncertainty shock makes agents distrust the mapping from aggregate events to personal outcomes.
- The model predicts safe-asset demand, capital immobility, liquidity hoarding, and market disengagement even without an explicit run equilibrium.

Interpretation: The model identifies a crisis mechanism that can coexist with, but is distinct from, standard fire-sale mechanisms.

5.2 Why aggregate probabilities matter

The model assumes aggregate probabilities are known but individual incidence probabilities are Knightian. This distinction is central because the central bank's aggregation removes idiosyncratic distortions.

Interpretation: If agents were uncertain about aggregate probabilities in the same way as the central bank, the policy argument would be weaker. The contribution comes from the difference between individual robust decision making and collective feasibility.

5.3 Why complete markets do not eliminate the problem

The model permits trade in contingent claims and observable, contractible shocks. Yet robust preferences still distort the allocation.

Interpretation: This is a strong result because it shows that the inefficiency does not rely on missing insurance markets or hidden information. It arises from Knightian probability distortions.

5.4 Why no multiple equilibrium is needed

The paper does not require a Diamond-Dybvig style bad equilibrium. Agents are not panicking irrationally. They optimize under robust preferences.

Interpretation: Flight-to-quality behavior can be individually rational and equilibrium consistent while still socially inefficient.

5.5 Empirical predictions and possible tests

The model suggests looking for:

- flight-to-quality behavior after new and poorly understood events;
- cross-market liquidity declines even in markets not directly hit by the initial shock;
- hoarding or drawdowns of credit lines when agents fear being late in a liquidity queue;
- a large effect of credible central-bank backstops even when little public money is actually disbursed;
- a difference between repeated, well-understood shocks and novel shocks of similar size.

Interpretation: A central empirical implication is that novelty matters. Similar losses may have very different macro-financial consequences depending on whether the loss causes agents to distrust their models.

6 Section-by-section study guide

6.1 Introduction

The introduction motivates the distinction between risk and uncertainty. It uses episodes such as Penn Central, 1987, LTCM, and 9/11 to argue that extreme events trigger model reassessment and worst-case thinking, not only higher variance.

Study tip. Focus on why the authors argue that flight-to-quality episodes are often one-of-a-kind. This motivates Knightian uncertainty rather than ordinary risk aversion.

6.2 Section I: The Model

This section sets up the shock tree, the benchmark allocation, the robust allocation, and implementations through financial intermediation or contingent claims.

Study tip. The key comparison is $K = 0$ versus $K > 0$. In the benchmark, scarce liquidity is allocated more to first-wave shocks. Under robustness, agents raise their demand for late-state insurance.

6.3 Section II: Collective Bias and Intervention

This section introduces the central bank objective and shows that individual robust allocations are collectively biased. The central bank can improve outcomes by reallocating liquidity from second-shock insurance toward first-shock insurance.

Study tip. The central bank is not assumed to know more. The difference is that it aggregates over agents, while each agent evaluates a private worst case.

6.4 Section III: Lender of Last Resort

This section translates the abstract reallocation into a feasible policy. A credible promise to intervene in extreme states reduces agents' worst-case anxiety and leads them to allocate private liquidity more efficiently.

Study tip. Remember that the policy's total value exceeds its direct value because of private portfolio reoptimization.

6.5 Final Remarks and Appendix

The final remarks compare the Knightian uncertainty model to liquidation externality models and discuss financial innovation. The appendix formalizes the event tree and proves the propositions.

Study tip. The final section is important for policy interpretation: not every large shock warrants intervention. Intervention is valuable when the shock creates uncertainty and liquidity hoarding.

7 Figures and theoretical result blocks

7.1 Figure 1: Benchmark case + Figure 2: Robustness case

Read:

- **Figure 1** shows the benchmark event tree when $K = 0$. The economy can have no shocks, one wave, or two waves.
- In Figure 1, agents have precise probabilities. The probability of being hit first in a two-wave state and hit second in a two-wave state is symmetric.
- The consumption bundles C^s indicate what the planner gives an agent in each state.
- **Figure 2** keeps the same physical event tree but replaces objective probabilities with agent-specific worst-case probabilities.
- The robust agent increases the perceived probability of being second and lowers the perceived relative weight on being first.
- This probability distortion is the diagrammatic source of the flight-to-quality allocation.

Detailed interpretation: Figure 1 is the efficient-risk-sharing benchmark. Since second shocks are less likely than first shocks, a liquidity-scarce planner allocates more consumption to first-wave agents. Figure 2 changes only beliefs, not resources or technology. The event tree is the same, but the agent evaluates it through worst-case subjective probabilities. This makes the agent demand more protection for late liquidity needs, even though not all agents can be late.

Economic message: The contrast between the two figures is the whole paper in miniature: a small change from risk to uncertainty transforms contingent cross-insurance into precautionary self-insurance.

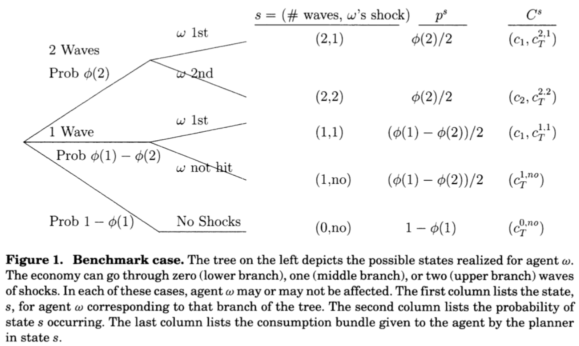


Figure 1: Benchmark case

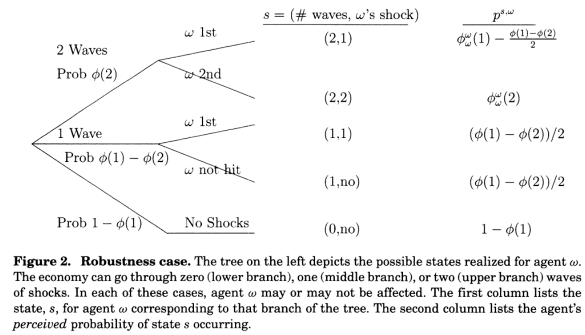


Figure 2: Robustness case

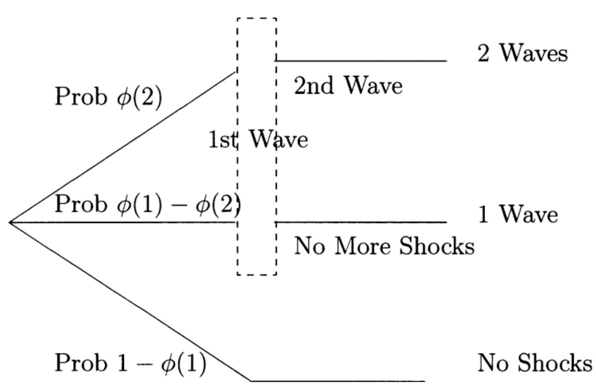
7.2 Appendix event tree: Event Tree and Probabilities

Read:

- The appendix event tree separates the aggregate number of waves from the agent's individual position in the shock sequence.
- The dashed box around the first wave highlights the ambiguity: an agent does not know whether the first wave is part of a one-wave event or the beginning of a two-wave event.
- The tree helps derive the individual probabilities $\phi_\omega(1)$ and $\phi_\omega(2)$ from the aggregate probabilities $\phi(1)$ and $\phi(2)$ plus the individual uncertainty parameter.

Detailed interpretation: The appendix makes precise why agents can be certain about the total probability of receiving a shock while being uncertain about whether they will be hit first or second. That structure is crucial: aggregate probabilities can be known, but individual ranking in the liquidity queue can remain ambiguous.

Economic message: The appendix figure clarifies why the central bank's aggregate objective differs from private max-min objectives. Individual queue-position fears average out across agents.



Appendix event tree and probabilities

7.3 Result block: Proposition 1

Read:

- If aggregate liquidity is insufficient, first-wave agents receive more liquidity than second-wave agents.
- If aggregate liquidity is sufficient, both waves can be fully insured.

Detailed interpretation: Proposition 1 shows what efficient insurance looks like before introducing Knightian uncertainty. It demonstrates that cross-insurance is valuable and that the socially efficient allocation is asymmetric across shock timing when liquidity is scarce.

Economic message: This establishes the benchmark against which flight-to-quality behavior is inefficient.

7.4 Result block: Proposition 2

Read:

- For small $K > 0$, agents partially shift insurance from first shocks toward second shocks.
- For large K, agents fully robustify and set $c_1 = Z = c_2$.
- The robust allocation collapses cross-insurance because agents fear late liquidity needs.

Detailed interpretation: Proposition 2 is the formal model of flight to quality. It converts a rise in Knightian uncertainty into hoarding and capital immobility. As K rises, the allocation moves away from socially efficient cross-insurance and toward individual liquidity preservation.

Economic message: The proposition explains why financial specialists can become liquidity demanders during crises rather than liquidity suppliers.

7.5 Result block: Proposition 3

Read:

- When $K > 0$ and liquidity is scarce, the robust allocation is biased toward second-shock insurance.
- A central bank concerned with aggregate ex post utility can improve outcomes by reallocating liquidity toward first-shock insurance.

Detailed interpretation: The central bank's improvement does not come from superior information. It comes from averaging across agents. Individual worst-case distortions cannot all be true simultaneously, so the aggregate allocation is excessively conservative.

Economic message: This is the paper's collective risk-management externality: individually prudent behavior is collectively wasteful.

7.6 Result block: Proposition 4

Read:

- A lender-of-last-resort policy that insures extreme tail states has value greater than the direct resources it contributes.

- The extra value comes from private portfolio reoptimization.

Detailed interpretation: Private agents reduce wasteful tail self-insurance when they trust that the central bank can provide resources in the extreme state. They then use private liquidity to insure more likely states, which increases social value even if the central bank rarely disburses resources.

Economic message: This is why credible central-bank backstops can stabilize markets without necessarily spending much public money.

8 Detailed result map

- **Risk-sharing benchmark:** Scarce liquidity should be allocated more to first waves than second waves.
- **Knightian uncertainty:** Agents overweight the possibility that they will be hit after others.
- **Flight to quality:** Robust private decisions become liquidity hoarding and capital immobility.
- **Collective bias:** Everyone cannot be late; individual worst-case beliefs do not aggregate into a feasible collective worst case.
- **Central bank role:** The central bank can reallocate risk management toward more likely shocks because it evaluates aggregate utility.
- **LLR complementarity:** Tail public insurance induces better private insurance of non-tail shocks.
- **Policy constraint:** The lender of last resort must be credible and truly last resort; early insurance can worsen distortions.

9 Short summary

- The paper studies flight-to-quality episodes driven by Knightian uncertainty and liquidity shortages.
- Agents know aggregate probabilities but are uncertain about their own position in the sequence of liquidity shocks.
- With precise probabilities and scarce liquidity, efficient cross-insurance gives more resources to first-wave shocks.
- Under Knightian uncertainty, agents worry about being hit late and demand extra late-state protection.
- As uncertainty rises, agents move from partial cross-insurance toward self-insurance.
- This behavior maps into liquidity hoarding, safe-asset demand, and immobility of risk capital.
- The social cost is that agents collectively guard against an impossible configuration of private worst cases.
- A central bank can improve aggregate outcomes by recognizing that individual fears average out.
- A lender-of-last-resort promise in extreme states can reduce private anxiety and unlock private liquidity.
- Public insurance and private insurance are complements in this environment, not substitutes.
- The policy must be reserved for tail states; early interventions can create moral-hazard-like distortions.
- The paper distinguishes uncertainty shocks from liquidation shocks and multiple-equilibrium runs.

10 Conclusion

10.1 Core conclusion

Caballero and Krishnamurthy argue that severe flight-to-quality episodes are best understood as crises of Knightian uncertainty interacting with liquidity scarcity. When agents distrust their models, they protect themselves against worst-case personal liquidity positions. These individually prudent actions create a collectively inefficient allocation of scarce liquidity. The central bank can improve outcomes not because it knows more than private agents, but because it evaluates the aggregate allocation rather than each agent's private worst case.

10.2 Economic intuition

The intuition is that crisis-time liquidity has a queue-like structure. Agents do not know whether they will need liquidity early or late. If they fear being late, they try to reserve liquidity for themselves. But if everyone reserves liquidity for the possibility of being late, too little liquidity is made available for the more likely early shocks. The economy becomes excessively defensive. This is the model's version of a flight to quality: agents abandon contingent risk sharing and demand safe liquid claims.

10.3 Incremental contribution to crisis theory

The paper's incremental contribution is to add a new reason for crises to be severe. In liquidation-externality models, prices fall because assets must be sold or collateral is scarce. In run models, bad equilibria arise because agents coordinate on withdrawals. In this model, even with complete contingent claims and observable shocks, agents' robust beliefs create excessive self-insurance. The crisis can therefore arise from uncertainty about the environment itself.

10.4 Policy implications

The model supports lender-of-last-resort policies, but in a very specific form. The central bank should provide credible insurance against rare tail states, not routine insurance against common losses. Such a promise reduces the need for private agents to hoard liquidity against those tail states and frees private liquidity for more likely shocks. The model therefore rationalizes why a central-bank commitment can have large effects even if little money is ultimately spent.

10.5 Why public and private insurance can be complements

A distinctive feature of the model is that public insurance can increase the efficiency of private insurance. When agents over-insure the wrong states, public backstops in those states make them rebalance toward better private insurance. This differs from standard moral hazard, in which public insurance crowds out private protection. Here, the public policy works partly by changing the composition of private risk management.

10.6 Financial innovation and novel shocks

The paper also explains why crises are often associated with new financial instruments or new market structures. Novelty makes it difficult for agents to rely on historical probability models. A shock to a new market may therefore generate Knightian uncertainty, even if a similar-sized shock to an old market would not. This provides a way to understand why some large shocks do not cause crises while some smaller but unfamiliar shocks do.

10.7 Limitations

The paper is theoretical and does not estimate the magnitude of K, the size of liquidity shortages, or the empirical value of lender-of-last-resort commitments. It abstracts from fire-sale price dynamics, endogenous leverage, strategic default, and explicit institutional features of modern balance sheets. In real crises, uncertainty shocks likely interact with collateral constraints, fire sales, and coordination failures. The model should therefore be read as isolating one mechanism, not as a complete crisis theory.

10.8 Takeaway

The takeaway is that financial crises can be caused not only by losses, leverage, or runs, but also by a collapse of confidence in the model used to interpret the environment. When uncertainty makes everyone demand safety at once, private risk management becomes socially excessive. A credible last-resort backstop can reduce that collective overprotection and restore the private sector's willingness to use scarce liquidity productively.